Pattern Recognition 165 (2025) 111592

Contents lists available at ScienceDirect

Pattern Recognition

journal homepage: www.elsevier.com/locate/pr

DSDC-NET: Semi-supervised superficial OCTA vessel segmentation for false positive reduction

Xinyi Liu, Hailan Shen, Wenyan Zhong, Wanqing Xiong, Zailiang Chen *

School of Computer Science and Engineering, Central South University, Changsha, 410083, China

ARTICLE INFO

Keywords:
False positive reduction
Vessel segmentation
Semi-supervised learning
Topological features
Spatial information

ABSTRACT

Accurate vessel segmentation in Optical Coherence Tomography Angiography (OCTA) is essential for ocular disease diagnosis, monitoring, and treatment assessment. However, most current automatic segmentation methods overlook false positives in the segmentation results, leading to potential misdiagnosis and delayed treatment. To address this issue, we propose a Dynamic Spatial Semi-Supervised Vessel Segmentation with Dual Topological Consistency (DSDC-NET) for retinal superficial OCTA images. The network integrates a Dynamic Spatial Attention Mechanism that combines snake-shaped convolution, which captures tubular fine structures, with spatial attention to suppress background noise and artefacts. This design enhances vessel region responses while accurately capturing complex local structures, thereby reducing false positives arising from inaccurate localisation of vessel details. Furthermore, Dual Topological Consistency Loss integrates the Persistent Homology features of the vessel system with the topological skeleton features of major vessels, enhancing branching pattern recognition. A Warm-up mechanism balances the focus of the network between major and branch vessels across training phases, mitigating false positives from inadequate branching structure learning. Comprehensive evaluations on ROSE-1, OCTA-500, and ROSSA datasets demonstrate the superiority of DSDC-NET over existing methods. Notably, DSDC-NET effectively reduces the false discovery rate and improves segmentation accuracy, validating its effectiveness in reducing false positives.

1. Introduction

Optical Coherence Tomography Angiography (OCTA) is a non-invasive imaging technique that visualises retinal microvascular structures, aiding in the diagnosis of diabetic retinopathy, age-related macular degeneration, and glaucoma [1]. Accurate vessel segmentation is crucial for these diagnostic tasks. However, the high resolution of OCTA images requires significant time and effort for manual annotation [2], and high costs limit annotation availability. Consequently, precise automatic segmentation is crucial to improving diagnostic efficiency and accuracy. Automated methods enable faster identification of vessel lesions, supporting timely treatment. However, OCTA segmentation networks are prone to false positives, misidentifying noise or non-vessel structures as vessels, as shown in Fig. 1(c). False positives arise primarily from two factors. Firstly, noise and artefacts interfere with vessel localisation, leading to misidentification of non-vessel regions. Secondly, insufficient learning of vessel branching patterns causes non-vessel extensions from incorrect branch points [3]. False positives hinder clinical diagnosis, complicate subsequent analysis, and affect disease assessment and treatment.

Examining the OCTA images in Fig. 1, it is evident that OCTA vessels are slender and intricate, particularly the branch vessels, which are fine and difficult to distinguish. Moreover, artefacts further reduce the contrast between vessel and non-vessel regions, complicating vessel localisation. Additionally, the vessel system consists of interconnected vessels, rather than isolated pixels or lines, placing greater demands on the segmentation to capture vessel orientation and adjacency relationships [4]. A network addressing false positives must accurately process vessel details, including local deformations and complex structures [5], while learning spatial relationships and suppressing background noise and artefacts. However, most existing studies [6] prioritise local features and vessel morphology, often neglecting spatial relationships within the vessel system. Integrating spatial data with local vessel features enhances localisation, improves network response, suppresses noise, and captures vessel structures accurately, reducing isolated points and broken segments, and mitigating false positives. Despite its importance, combining vessel details with spatial information to reduce false positives remains underexplored in current research.

* Corresponding author.
E-mail addresses: lxy_csu_edu@163.com (X. Liu), hn_shl@126.com (H. Shen), zzzwy228@163.com (W. Zhong), xiongwanqing@csu.edu.cn (W. Xiong), xxxyczl@csu.edu.cn (Z. Chen).

https://doi.org/10.1016/j.patcog.2025.111592
Received 11 October 2024; Received in revised form 4 March 2025; Accepted 10 March 2025
Available online 20 March 2025

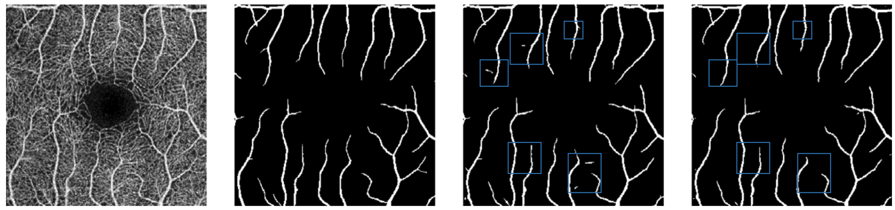


Fig. 1. An example illustrating the challenges faced by current OCTA vessel segmentation methods. (a) Original OCTA image, (b) Ground truth, (c) Results from the current state-of-the-art semi-supervised method HAIC-NET, and (d) Results of our proposed network.

In the OCTA vessel system, the hierarchical structure describes the arrangement from major vessels to branches, while the branching pattern denotes their distribution, forming a complete circulatory system. Both aspects underscore the need for accurately segmenting major vessels in HAIC-NET, the state-of-the-art method [7]: (1) From the hierarchical structure, major vessels serve as primary conduits for blood flow, influencing transport pathways and systemic functions. (2) From the branching pattern, accurate segmentation of major vessels identifies key branch points critical for blood flow distribution and velocity [8]. Given the two points, accurate segmentation of major vessels is vital for delineating the entire OCTA vessel network. Although HAIC-NET performs well, it still has certain limitations. It prioritises the topological connectivity of major vessels but overlooks challenging branch vessels and fails to capture higher-dimensional relationships in complex tubular structures, impacting segmentation accuracy. As shown in Fig. 1(c), the model struggles with small, curved features, causing incorrect segmentation. Insufficient topological learning hinders vessel branch point identification, leading to false positives. Thus, reducing false positives requires broader branch vessel consideration while preserving major vessel segmentation accuracy. Additionally, the vessel connectivity and branching patterns must be considered.

Based on the above analysis, the Dynamic Spatial Semi-Supervised Vessel Segmentation with Dual Topological Consistency (DSDC-NET) is proposed to address false positives in OCTA vessel segmentation. The network consists of two main components: the Dynamic Spatial Attention Mechanism (DSAM) and the Dual Topological Consistency Loss (DTC). DSAM combines snake-shaped convolutions [5] with spatial attention, adapting to geometric deformations and complex vessel structures while enhancing vessel region responses and suppressing background noise for precise vessel localisation. DTC includes the topological skeleton loss L_{TS} and segmentation loss L_{Seg}, evaluating segmentation accuracy of major and branch vessels using topological skeletons and Persistent Homology. A Warm-up mechanism adjusts loss preferences during different training stages, enabling the network to learn branching patterns and reduce false positives. To validate the effectiveness of our network, we performed experiments on three public datasets: ROSE-1, OCTA-500, and ROSSA. The main contributions of this study are as follows:

- To reduce false positives in OCTA vessel segmentation, a Dynamic Spatial Semi-Supervised Vessel Segmentation with Dual Topological Consistency is proposed for retinal OCTA images.
- To tackle false positives from inaccurate vessel localisation, we propose the Dynamic Spatial Attention Mechanism. Integrating snake-shaped convolution with spatial attention enhances the ability to capture vessel details while suppressing background noise, thereby reducing false positives.
- To mitigate false positives from inadequate vessel branching pattern learning, we propose Dual Topological Consistency Loss, combining topological skeleton and Persistent Homology. A Warm-up mechanism adapts learning preferences across training phases, thereby reducing false positives.
- We validate DSDC-NET through experiments on three publicly available datasets (ROSE-1, OCTA-500, and ROSSA). Ablation studies confirm the practicality of the network. Results show that DSDC-NET outperforms current state-of-the-art methods and is comparable to some fully supervised methods from recent years.

The remainder of this paper is structured as follows. Section 2 provides an overview of the background and current research in vessel segmentation, focusing on fully and semi-supervised methods. Section 3 outlines the training procedure for DSDC-NET, the Dynamic Spatial Attention Mechanism, and Dual Topological Consistency Loss. Section 4 presents experimental results and visualisations validating the effectiveness of DSDC-NET, with ablation studies confirming its practicality. Finally, Section 5 concludes our work, discusses our limitations, and suggests potential future directions.

2. Related work on vessel segmentation

2.1. Fully supervised vessel segmentation methods

Early vessel segmentation primarily relied on fully supervised methods. Since 2015, networks like U-NET [9] have formed the basis for automatic OCTA vessel segmentation. OCTA-NET [10] proposes a coarse-to-fine segmentation approach, while ACRROSS [11] improves the precision with disentanglement techniques. H2C-Net [12] enhances segmentation by leveraging image depth and soft-layered features within OCTA volume data. However, its complexity and need for precise annotations limit its broader applicability. Significantly, the lack of high-quality labeled data prevents models from learning key features, impairing performance. This causes failure in capturing branch connectivity and geometry, misidentifying non-vessel structures as vessels, leading to false positives. Consequently, the reliance on labeled data limits the effectiveness of fully supervised methods for OCTA vessel segmentation.

2.2. Semi-supervised vessel segmentation methods

Semi-supervised methods mitigate the scarcity of labeled data by combining labeled and unlabeled images, balancing accuracy with the richness of unlabeled data to enhance model performance [13]. Recent methods for vessel segmentation have emerged, inspired by classical semi-supervised approaches like Mean Teacher [14] and MixMatch [13]. Li et al. [15] propose a hierarchical deep network with an attention mechanism to localise low-contrast capillary regions. While effective in validating the attention mechanism, it does not address complex vessel branching. EXP-Net [16] improves weak vessel connectivity through knowledge enhancement and connectivity modules. However, it still struggles with accurate vessel connectivity due to artefacts in OCTA images, leading to lost details and incorrect segmentation. Self-supervised methods [17] use proxy tasks to leverage large volumes of unlabeled data to learn inherent structures. Yet, challenges like high vessel density, terminal vessel segmentation, tissue interference, and complex lesion patterns in OCTA images hinder their direct application and necessitate refinement. Consequently, approaches combining self-supervised tasks with semi-supervised strategies have emerged. DCSS-NET [6] integrates a jigsaw puzzle task with dual-consistency regularisation, achieving notable performance, but the jigsaw task may disrupt topological features crucial for connectivity. Shen et al. [7] proposed HAIC-NET in 2024, combining homology classification with data perturbation and topological consistency to enhance vessel feature extraction from unlabeled data. However, it lacks awareness of global topological structures, leading to inadequate learning of vessel branching patterns, and causing false positives. Furthermore, since the self-supervised task is trained without annotated data, the network may incorrectly identify non-vessel regions as vessels without accurate supervisory signals, a problem exacerbated by noise and artefacts in OCTA.

3. Methodology

3.1. Network architecture

To alleviate the phenomenon of false positives encountered in OCTA vessel segmentation, a Dynamic Spatial Semi-Supervised Vessel Segmentation with Dual Topological Consistency is proposed for retinal OCTA images. Fig. 2 illustrates the architecture of the proposed DSDC-NET architecture.

For a given set of labeled images $S_L = \{x_l, y_l\}_{l=1}^{M}$ and unlabeled images $S_U = \{x_u\}_{u=1}^{N}$, where M represents the number of labeled images, N represents the number of unlabeled images, $M \ll N$, $S = S_L \cup S_U$, we apply image augmentation techniques to S such as horizontal flipping, vertical flipping, rotation, and random cropping, collectively referred to as operation A. The original and augmented images are fed into a shared encoder E, and the output of E is processed using the Dynamic Spatial Attention Mechanism, as shown in Fig. 2(a). The processed results of the augmented images are fed into a main decoder D_{ma}, while the processed results of the original images are fed into an auxiliary decoder D_{au}. The output of D_{au} is processed using operation A, and the topological skeleton loss L_{TS} is calculated between the result and the output of D_{ma}. Additionally, segmentation loss between the output of the labeled images and the ground truth is computed as L_{Seg}. Together, L_{TS} and L_{Seg} form the Dual Topological Consistency Loss, as shown in Fig. 2(b). DSDC-NET includes a Dynamic Spatial Attention Mechanism and a Dual Topological Consistency Loss, and detailed explanations will be provided in Sections 3.2 and 3.3.

3.2. Dynamic Spatial Attention Mechanism

The Dynamic Spatial Attention Mechanism integrates snake-shaped convolution and spatial attention blocks. Snake-shaped convolution captures vessel details through dynamic structures but is vulnerable to background noise in OCTA images. Conversely, spatial attention distinguishes foreground from background effectively but struggles with complex vessel details. To address this, we creatively combine the strengths of two components, resolving inaccurate vessel detail localisation that leads to false positives. This integration suppresses background noise while preserving vessel details, enabling DSDC-NET to balance global and local information. The spatial attention mechanism strengthens the vessel region response, while snake-shaped convolution ensures the accuracy of local vessel details, enabling the network to localise vessel details accurately and, as a result, reduce the false positive rate. Section 3.2.1 will present the existing snake-shaped convolution network, and Section 3.2.2 will detail the proposed Dynamic Spatial Attention Mechanism.

3.2.1. Snake-shaped convolution network

The snake-shaped convolution network [5] employs dynamic structures to capture elongated tubular shapes, enhancing critical feature perception. It integrates deformable offsets for improved kernel flexibility, focusing on complex geometry. To prevent perceptual field deviation from unconstrained offsets learning, an iterative strategy selects positions sequentially, ensuring continuity and limiting field expansion. Unlike traditional methods, it linearises standard kernels along the x and y axes. For example, with a convolution kernel of size 9, each grid position in the x-axis direction is denoted as $K_{i\pm c} = (x_{i\pm c}, y_{i\pm c})$, where $c = 0, 1, 2, 3, 4$ represents the horizontal distance from the centre grid position. The selection of each network position $K_{i\pm c}$ in the convolution kernel K is an accumulated process, starting from the central position K_i, with each grid position determined by adding an offset Δ. Specifically, the variation of the convolution kernel along the x-axis is:

$$K_{i\pm c} = \begin{cases} (x_{i+c}, y_{i+c}) = \left(x_i + c, y_i + \sum_i^{i+c} \Delta y\right), \\ (x_{i-c}, y_{i-c}) = \left(x_i - c, y_i + \sum_{i-c}^{i} \Delta y\right), \end{cases} \quad (1)$$

the variation of the convolution kernel along the y-axis is:

$$K_{j\pm c} = \begin{cases} (x_{j+c}, y_{j+c}) = \left(x_j + \sum_j^{j+c} \Delta x, y_j + c\right), \\ (x_{j-c}, y_{j-c}) = \left(x_j + \sum_{j-c}^{j} \Delta x, y_j - c\right), \end{cases} \quad (2)$$

Since the offset Δ is typically a decimal, while coordinates are integers, the snake-shaped convolution network uses bilinear interpolation to represent decimal positions. The bilinear interpolation kernel B is decomposed into two one-dimensional kernels for a two-dimensional transformation. This design ensures the receptive field size is retained during deformation, allowing better adaptation to elongated tubular structures and more accurate feature perception.

3.2.2. Training with Dynamic Spatial Attention Mechanism

Although the snake-shaped convolution network uses deformable offsets Δ for flexibility, this may cause the kernel to adapt to noise or irrelevant features, misidentifying them as part of the target. Additionally, bilinear interpolation used to handle decimal offsets and integer coordinates may introduce errors, particularly at image edges or in complex textures, causing the kernel to misinterpret certain regions and resulting in false positives. Given the complex morphology of vessels and lesions in OCTA images, relying solely on the snake-shaped convolution may result in kernel misjudgemnts and false positives. In contrast, the Spatial Attention Mechanism [18] adaptively focuses on vessel regions, improving shape and position assessment, enhancing responses in key areas, and suppressing background noise and artefacts, which the snake-shaped convolution struggles with. Thus, the Dynamic Spatial Attention Mechanism combines spatial attention, which effectively mines spatial information, with snake-shaped convolution, adept at handling locally meandering structures, to balance both local and global information, addressing false positives from inaccurate vessel detail localisation and improving network performance, as shown in Fig. 2(a).

For the output of the encoder $\hat{x}$, we input it simultaneously into both the spatial attention module and the snake-shaped convolution module to obtain the spatial feature component x_{sp} and the snake-shaped convolution feature component x_{sn}. Specifically, in the spatial attention module, $\hat{x}$ is processed as described in Eq. (3) to calculate the attention similarity ω_{af} and extract features:

$$\omega_{af} = softmax\left(W_Q(\hat{x}) \otimes W_K(\hat{x})\right) \quad (3)$$

where W_Q represents query vector extraction, W_K represents key vector extraction, $\otimes$ represents element-wise multiplication, and $softmax(\cdot)$ represents the softmax operation.

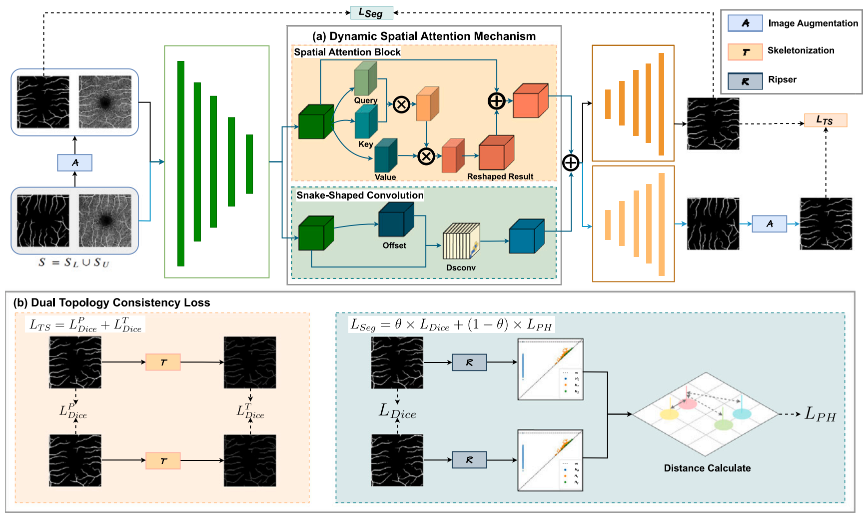


Fig. 2. The overview of the proposed DSDC-NET architecture. DSDC-NET comprises two main components: (a) Dynamic Spatial Attention Mechanism, enhancing the capacity to extract spatial information while focusing on elongated and curved local features, thus reducing false positives; and (b) Dual Topological Consistency Loss, improving the network proficiency in learning the structural hierarchy and branching pattern of vessel system, increasing the likelihood of accurate major vessel segmentation and mitigating false positives from insufficient structural learning. We use three lines, each ranging in colour from dark to light, to distinguish the data flow of the enhanced image, the whole image, and the original image.

Subsequently, the obtained attention similarity ω_{af} is processed as shown in Eq. (4) to calculate the spatial offset component $\mathbf{x}'_{sp}$:

$$\mathbf{x}'_{sp} = \omega_{af} \otimes W_V(\hat{\mathbf{x}}) \tag{4}$$

where W_V represents value vector extraction.

Finally, the spatial offset component $\mathbf{x}'_{sp}$ is projected to have the same shape as $\hat{x}$, and processed as shown in Eq. (5) to obtain the spatial feature component x_{sp}:

$$x_{sp} = \Psi\left(x'_{sp}\right) \oplus \hat{x} \tag{5}$$

where $\Psi(\cdot)$ represents the projection operation, and $\oplus$ represents element-wise addition.

In the snake-shaped convolution module, we first use Eq. (1) and (2) to generate the snake-shaped convolution Dsconv for OCTA images. Subsequently, we process it as shown in Eq. (6) to obtain the snake-shaped convolution feature component x_{sn}:

$$x_{sn} = Dsconv\left(\hat{x}, M_{off}(\hat{x})\right) \tag{6}$$

where $M_{off}(\cdot)$ represents the offset convolution calculation.

Finally, for the spatial feature component x_{sp} and the snake-shaped convolution feature component x_{sn}, we use Eq. (7) to compute the output of the DSAM $\breve{x}$. This output will be used as the input for both D_{ma} and D_{au}:

$$\breve{x} = x_{sp} + x_{sn} \tag{7}$$

3.3. Dual topological consistency loss

3.3.1. OCTA topological structure

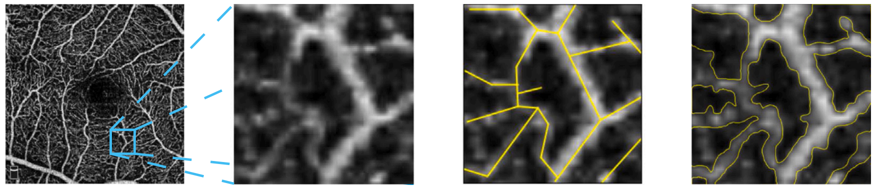
With the widespread use of image enhancement techniques, topological features [19] have become vital for analysing complex image data due to their robustness in shape and structure. In OCTA images, topological features such as connected components, nodes, edges, loops, vessel density, and connectivity quantify the global vessel structure [20], revealing physiological and pathological mechanisms that support diagnosis, screening, and treatment evaluation. Persistent Homology, a Topological Data Analysis (TDA) method, tracks the evolution of topological features across scales, providing valuable insights for OCTA vessel segmentation. Taking a part of an OCTA image marked by the blue box in Fig. 3(a) as an example, Persistent Homology in OCTA images typically includes three data dimensions: the connected components of the vessel system, denoted as H_0 and annotated by yellow lines in Fig. 3(b), reflecting vessels connectivity; one-dimensional holes or loops, denoted as H_1 and annotated by yellow closed curves in Fig. 3(c), indicating vessel redundancy and robustness; and the cavities in the vessel system, denoted as H_2, which cannot be shown in two dimensions. The H_2 dimension reflects potential vessel damage and the development of other lesions, offering significant clinical relevance compared to other dimensions.

Major vessels in the OCTA vessel system are sparse but thicker, forming the foundation for numerous thinner branches. The success of HAIC-NET [7] has underscored the importance of accurately segmenting major vessels to ensure correct branch vessel identification. While Persistent Homology provides a comprehensive understanding of vessel morphology, it overlooks the substantial differences between major and branch vessels. The consistency assessment of the topological skeleton mitigates segmentation errors in major vessels, addressing the limitations of Persistent Homology. Accordingly, the Dual Topological Consistency Loss is designed to balance the topological structure requirements of both major and branch vessels during segmentation.

3.3.2. Training with Dual Topological Consistency Loss

We design the Dual Topological Consistency Loss, as illustrated in Fig. 2(b), which ensures the network preserves the connectivity and morphological accuracy of the major vessels while enhancing the detection of branch points, thereby reducing the risk of false positives.

We use two topological features to assist us in loss construction. For OCTA images, we utilise the Ripser to perform Persistent Homology feature extraction. Specifically, we first convert the image into a point cloud, where each point is characterised by its grey value. Then, we construct the topological space by applying a filtration process with a progressively increasing distance threshold between points. This process begins at zero, with the distance thresholds between connected points gradually increasing until all points are fully connected at a specific scale, forming a simplicial complex. By analysing the homotopy clusters of the complex across different threshold ranges, we can identify the topological features that persist throughout the image. For segmentation results, we employ a centreline skeleton extraction method to derive the topological skeleton [21]. This involves performing morphological operations on the image, such as erosion and dilation, to refine and extract the skeleton structure.



(a) (b) (c)

Fig. 3. OCTA Topological Structure. (a) OCTA image with an example marked by a blue box, (b) connected components H_0 of the vessel system, (c) one-dimensional holes or loops H_1 in the vessel system.

The topological skeleton loss L_{TS} is calculated based on the main decoder output p_m and the image enhancement results from the auxiliary decoder output p_a. For labeled data, the segmentation loss L_{Seg} between the ground truth y_l and the output of the main decoder p_m is additionally calculated. Specifically, for both unlabeled data D_U and labeled data D_L, the Dice prediction loss L^P_{Dice} between p_m and p_a is calculated using Eq. (8):

$$L^P_{Dice} = Dice\left(p_m, p_a\right) \tag{8}$$

where $Dice(\cdot,\cdot)$ represents the Dice loss calculation. The Dice skeleton topological loss L^T_{Dice} between these skeletons is calculated using Eq. (9):

$$L^T_{Dice} = Dice\left(T\left(p_m\right), T\left(p_a\right)\right) \tag{9}$$

where $T(\cdot)$ represents the extraction of the topological skeleton. Finally, we use Eq. (10) to calculate the topological skeleton loss L_{TS}:

$$L_{TS} = L^P_{Dice} + L^T_{Dice} \tag{10}$$

For labeled data, we use Eq. (11) to calculate the Dice loss L_{Dice} between y_l and p_m:

$$L_{Dice} = Dice\left(y_l, p_m\right) \tag{11}$$

Subsequently, we use Persistent Homology to compute the topological data T^{PH}_y and T^{PH}_p for y_l and p_m in three dimensions, where $T^{PH}_y, T^{PH}_p = \left\{H_0, H_1, H_2\right\}$. It then calculates the Wasserstein distance between the data in each dimension of T^{PH}_y and T^{PH}_p, where the Wasserstein distance is a metric used to measure the distance between two probability distributions. The results from the three dimensions are weighted and summed to obtain the Persistent Homology loss L_{PH}, as shown in Eq. (12):

$$\begin{aligned} L_{PH} = {} & \alpha \times \mathcal{D}_{wass}\left(T^{PH}_y\left[H_0\right], T^{PH}_p\left[H_0\right]\right) \\ & + \beta \times \mathcal{D}_{wass}\left(T^{PH}_y\left[H_1\right], T^{PH}_p\left[H_1\right]\right) \\ & + \gamma \times \mathcal{D}_{wass}\left(T^{PH}_y\left[H_2\right], T^{PH}_p\left[H_2\right]\right) \end{aligned} \tag{12}$$

where α, β, γ is the weight used to balance the results from the three dimensions, $\mathcal{D}_{wass}(\cdot)$ represents the Wasserstein distance calculation, $T^{PH}_y\left[H_i\right]$, $i = 0, 1, 2$ represents ith dimension result of T^{PH}_y, $T^{PH}_p\left[H_i\right]$, $i = 0, 1, 2$ represents ith dimension result of T^{PH}_p. A Warm-up mechanism is established to reduce interference from abnormal topological data caused by insufficient early training. During the early stages of training, L_{Dice} has a larger weight, allowing the model to focus more on the correctness and completeness of the segmentation results, thereby establishing an accurate foundation for the segmentation of major vessels; in the later stages, L_{PH} has a larger weight, prompting the model to concentrate on the accuracy of the Persistent Homology features of the segmentation results, thus ensuring the correct representation of the branching pattern. As shown in Eq. (13):

$$L_{Seg} = \theta \times L_{Dice} + (1 - \theta) \times L_{PH} \tag{13}$$

where θ is a coefficient related to the training epoch, and the relationship between the training epoch k and θ is described by Eq. (14):

$$\theta = 1 / \left(\sqrt{k}/\tau + 1\right) \tag{14}$$

where τ is the temperature coefficient.

Table 1
Definition of symbols used in Algorithm 1.

Symbol	Meaning	Symbol	Meaning
x	OCTA Image	y	Ground Truth
M	Number of Labeled Images	N	Number of Unlabeled Images
S_L	Labeled Dataset	S_U	Unlabeled Dataset
A	Image Augmentation	E	Encoder
$\hat{x}$	Output of the Encoder	$\check{x}$	Output of the DSAM
D_{ma}	Main Decoder	D_{au}	Auxiliary Decoder
p_m	Main Decoder Prediction	p_a	Auxiliary Decoder Prediction
L_{Seg}	Segmentation Loss	L_{TS}	Topological Skeleton Loss
L	Total Loss	ω	Ramp-up Coefficient

3.4. Loss function

DSDC-NET includes two stages: training with labeled data and training with unlabeled data. For labeled data, the network is trained using the ground truth and employs the segmentation loss L_{Seg} as the loss function. Additionally, to enhance the connectivity constraint of the major vessels, the topological skeleton loss L_{TS} is used as the loss function. The total loss function for labeled data L_S is represented by Eq. (15):

$$L_S = L_{Seg} + L^L_{TS} \tag{15}$$

where L^L_{TS} represents the topological skeleton loss of labeled data. For unlabeled data, we use the topological skeleton loss L_{TS} as a constraint function. The total loss function for unlabeled data L_U is represented by Eq. (16):

$$L_U = L^U_{TS} \tag{16}$$

where L^U_{TS} represents the topological skeleton loss of unlabeled data.

In summary, the total loss function for DSDC-NET is given by Eq. (17):

$$L = L_S + \omega \times L_U \tag{17}$$

where ω is the ramp-up coefficient for weighting the loss for training with unlabeled data. The relationship between ω and the training epoch k is given by Eq. (18):

$$\omega = 10 \times e^{-5 \times \left(1 - k/N_e\right)^2} \tag{18}$$

where N_e is the balancing coefficient.

Overall, the training process of DSDC-NET is illustrated in Algorithm 1.

Considering the extensive use of symbols in Algorithm 1, we provide the symbol definitions in Table 1.

4. Experiments

4.1. Datasets

To validate the segmentation performance of DSDC-NET on OCTA images, we conducted experiments on three public datasets (ROSE-1,

Algorithm 1 DSDC-NET Training Process

Require: Labeled OCTA Dataset $S_L = \{x_L, y_L\}_{l=1}^{M}$ and Unlabeled OCTA Dataset $S_U = \{x_U\}_{u=1}^{N}$
Ensure: Robust vessel segmentation network f
Apply image enhancement A to S_L, S_U to obtain S_L^A, S_U^A
for $i = 0$ to $epoch$ **do**
  $\hat{S}_L, \hat{S}_U, \hat{S}_L^A, \hat{S}_U^A = E\left(S_L, S_U, S_L^A, S_U^A\right)$
  $\check{S}_L, \check{S}_U, \check{S}_L^A, \check{S}_U^A = DSAM\left(\hat{S}_L, \hat{S}_U, \hat{S}_L^A, \hat{S}_U^A\right)$
  $p_{mL}^A, p_{mU}^A = D_{ma}\left(\check{S}_L^A, \check{S}_U^A\right)$
  $p_{aL}, p_{aU} = D_{au}\left(\check{S}_L, \check{S}_U\right)$
  Calculate the segmentation loss L_{Seg} between y_L^A and p_{mL}^A using Equation (13).
  Apply image enhancement A to p_{aL}, p_{aU} to obtain p_{aL}^A, p_{aU}^A
  Calculate the topological skeleton loss L_{TS}^L between p_{aL}^A and p_{mL}^A using Equation (10).
  Calculate the topological skeleton loss L_U between p_{aU}^A and p_{mU}^A using Equation (10).
  $L_S = L_{Seg} + L_{TS}^L$
  $L = L_S + \omega \times L_U$
  Perform the backward of L to update the parameters of f
end for
return f

OCTA-500, and ROSSA). Only a limited portion of the image annotations from these datasets are used for supervised training.

ROSE-1 [10], introduced by Ma et al. comprises 117 OCTA images from 39 subjects, including 26 Alzheimer's disease (AD) patients and 13 healthy controls. It features three subsets: superficial vascular complex (SVC), deep vascular complex (DVC), and combined SVC+DVC. All scans are obtained using the RTVue XR Avanti SD-OCT system with a resolution of 304 × 304 pixels. Our experiments use the SVC subset, with 30 training and 9 testing images. 6 training images are designated for validation, yielding 24 for training, 6 for validation, and 9 for testing.

OCTA-500 [22], introduced by Li et al. is a comprehensive dataset comprising 3D OCT and OCTA data from 500 eyes. It includes six types of projection images, four demographic labels, and seven segmentation labels. For OCTA vessel segmentation, it provides 3×3 mm (OCTA-3M) and 6×6 mm (OCTA-6M) subsets with manual annotations. In DSDC-NET, we use the OCTA-3M subset at a 304 × 304 resolution, selecting 3 images for supervised training, 57 for unsupervised training, 20 for validation, and 20 for testing, with all results reported on the test set.

ROSSA [23], released by Ning et al. in 2024, is a large-scale OCTA dataset comprising 918 images with pixel-level annotations. Of these, with a resolution of 320 × 320 pixels, the first 300 images are manually annotated, while the remaining 618 images are semi-automatically annotated using the fine-tuned Segment Anything Model (SAM) [24]. Our experiments use the first 100 manually annotated images, allocating 60 for training, 20 for validation, and 20 for testing.

4.2. Implementation details

We utilise the classic U-Net [9], a widely adopted network in medical image processing, as our backbone. All experiments were performed on an NVIDIA GeForce RTX 3050 using PyTorch.

All images are normalised using the mean and standard deviation of the dataset before input. The normalised images are then centre-cropped to resize them to 288 × 288 pixels uniformly. Following this, a series of random augmentation strategies, including horizontal flipping, vertical flipping, rotation, and random cropping, are applied. The initial learning rate is set to 0.005, with a weight decay of 1×10^{-5} and a momentum of 0.9. We use the Adam optimiser to train the model for 300 epochs. The weights of L_{PH} for each dimension are set to $\alpha = 0.1$, $\beta = 0.3$, and $\gamma = 0.6$. During training, the balance coefficient N_e is set to 400, and the temperature coefficient τ is set to 21. A detailed discussion of the temperature coefficient settings is provided in Section 4.5.

4.3. Experimental configurations

The performance of DSDC-NET is compared with seven state-of-the-art methods, including MT [14], MixMatch [13], TCSM [25], TCSMv2 [26], SLC-NET [27], DCSS-NET [6], and HAIC-NET [7]. Among these, MT [14] and MixMatch [13] are originally designed for classification tasks, while TCSM [25] is specifically developed for skin lesion segmentation. DCSS-NET [6] and HAIC-NET [7] are tailored for OCTA vessel segmentation, whereas the remaining methods are general medical image segmentation approaches. Notably, TCSMv2 [26] employs data perturbation techniques, while SLC-NET [27] leverages cross-style consistency. To comprehensively evaluate the performance of DSDC-NET across different datasets, we adopt five widely used metrics in medical image segmentation: Accuracy (ACC), Dice Coefficient (DSC), False Discovery Rate (FDR), G-Mean, and Jaccard Index (JAC).

4.4. Comparison with other SOTA methods

4.4.1. Experiments on the ROSE-1 dataset

We compare segmentation performance on ROSE-1, where all semi-supervised methods utilise only 1 labeled image (3.3%) and 23 unlabeled images, with 6 reserved for validation. Additionally, we introduce two fully supervised methods, ACRROSS [11] and OCTA-NET [10], trained on all 30 labeled images (100%), enabling a direct comparison and demonstrating that DSDC-NET performs comparably to fully supervised approaches. All methods adopt U-Net as the backbone.

Table 2 presents the quantitative results of the experiments, demonstrating that DSDC-NET achieves the best performance to date and closely approaches the performance of fully supervised methods. Specifically, DSDC-NET outperforms the current SOTA method HAIC-NET by 0.62% in JAC, 0.37% in ACC, 0.46% in DSC, and 2.69% in FDR. Although HAIC-NET also considers the topological features of major vessels, DSDC-NET attains higher segmentation accuracy by more effectively incorporating global vessel Persistent Homology features.

Fig. 4 presents the visual segmentation results of DSDC-NET and a portion of the results from the semi-supervised methods. Compared to other methods, DSDC-NET demonstrates superior segmentation ability for complex vessel structures, as shown in the blue box, and effectively reduces false positives in the segmentation results, as indicated in the green box, enhancing connectivity and minimising isolated points.

4.4.2. Experiments on the OCTA-500 dataset

We conduct experiments on the OCTA-500 dataset to evaluate DSDC-NET, comparing it with other semi-supervised methods using a consistent dataset split. As a benchmark, we include the fully supervised method DB-UNET [28]. Table 3 shows that DSDC-NET, with only 5% annotations, outperforms other semi-supervised methods, improving by 0.6% in JAC, 0.37% in DSC, and 2.05% in FDR compared to HAIC-NET. It also achieves performance close to DB-UNET, demonstrating its ability to reduce false positives and enhance segmentation accuracy by learning global and local vessel structures. Moreover, Fig. 5 illustrates the visual segmentation results of DSDC-NET on OCTA-500 alongside those of part of semi-supervised methods. The results indicate that DSDC-NET more effectively captures vessel branching structures and mitigates false positives compared to other methods.

Table 2
Quantitative results for ROSE-1 using only 3.3% manual annotations. In the Type column, F denotes fully supervised learning, utilising all annotations to train the model. S represents semi-supervised learning, which combines supervised and unsupervised training. The bolded text indicates the current optimal result.

Method	Type	JAC ↑	ACC ↑	DSC ↑	FDR ↓	G-Mean ↑
OCTA-NET [10] (2020)[a]	F	/	0.9182	0.7697	0.1775	0.8361
ACCROSS [11] (2022)[a]	F	/	0.9240	0.7880	0.1650	0.8520
MT [14] (2017)	S	0.5035	0.8734	0.6693	0.3129	0.7826
TCSM [25] (2018)	S	0.5735	0.9051	0.7276	0.1787	0.7950
MixMatch [13] (2019)	S	0.5782	0.9045	0.7315	0.1960	0.8039
TCSM-v2[26] (2020)	S	0.5882	0.9056	0.7395	0.2005	0.8131
DCSS-NET [6] (2022)	S	0.6186	0.9126	0.7627	0.1852	0.8307
SLC-NET [27] (2024)	S	0.5804	0.9027	0.7338	0.2241	0.8185
HAIC-NET [7] (2024)	S	0.6228	0.9132	0.7661	0.1904	**0.8358**
Ours	S	**0.6290**	**0.9169**	**0.7707**	**0.1635**	0.8319

[a] Indicates that the data is sourced directly from relevant prior work.

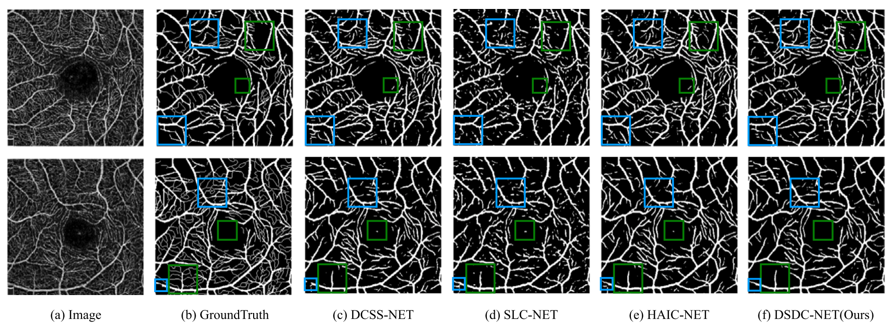


Fig. 4. Visualisation results on ROSE-1. The blue box indicates that DSDC-NET is more capable of segmenting complex vessels; the green box shows that DSDC-NET effectively reduces false positives, improving overall connectivity and reducing isolated pixel points in the OCTA image.

Table 3
Quantitative results for OCTA-500 using only 5% manual annotations.

Method	Type	JAC ↑	ACC ↑	DSC ↑	FDR ↓	G-Mean ↑
DB-UNet [28] (2023)	F	0.8287	0.9884	0.9061	0.1050	0.9548
MT [14] (2017)	S	0.7513	0.9827	0.8576	0.1394	0.9216
TCSM [25] (2018)	S	0.7785	0.9855	0.8749	**0.0810**	0.9125
MixMatch [13] (2019)	S	0.7600	0.9842	0.8632	0.0856	0.9028
TCSM-v2[26] (2020)	S	0.7843	0.9856	0.8788	0.0994	0.9238
DCSS-NET [6] (2022)	S	0.7973	0.9862	0.8869	0.1164	0.9406
SLC-NET [27] (2024)	S	0.7569	0.9832	0.8615	0.0977	0.9067
HAIC-NET [7] (2024)	S	0.8126	0.9869	0.8964	0.1039	**0.9447**
Ours	S	**0.8186**	**0.9876**	**0.9001**	0.0835	0.9387

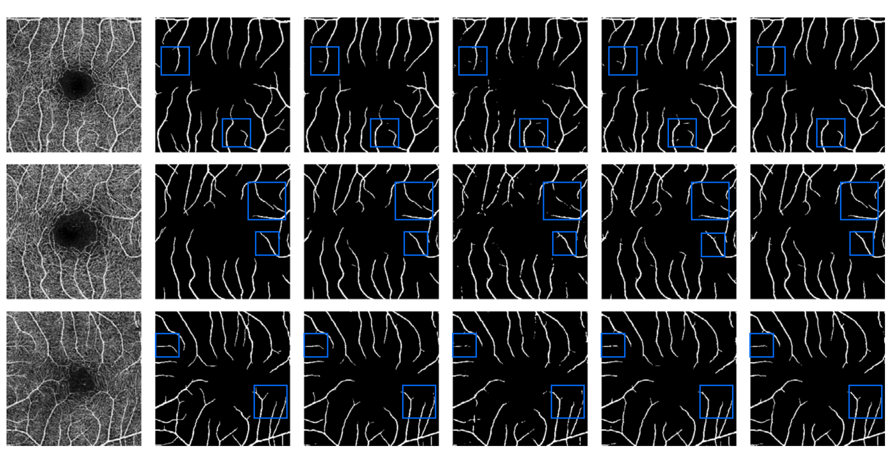


Fig. 5. Visualisation results on OCTA-500 with only 5% manual annotations. The blue box indicates where DSDC-NET outperforms other network segmentation.

4.4.3. *Experiments on the ROSSA dataset*

We also evaluate our network on the ROSSA dataset, comparing it with other semi-supervised methods. All experiments use the same dataset split. Table 4 shows the quantitative metrics of DSDC-NET on ROSSA. With only 5% annotation, DSDC-NET outperforms other semi-supervised methods, achieving improvements of 2.09% in JAC, 1.29%

Table 4
Quantitative results for ROSSA using only 5% manual annotations.

Method	JAC ↑	ACC ↑	DSC ↑	FDR ↓	G-Mean ↑
MT [14] (2017)	0.5454	0.9222	0.7009	0.3191	0.8417
TCSM [25] (2018)	0.7770	0.9747	0.8740	0.0787	0.9081
MixMatch [13] (2019)	0.7646	0.9716	0.8655	0.1135	0.9140
TCSM-v2[26] (2020)	0.7770	0.9743	0.8737	0.0757	0.9073
DCSS-NET [6] (2022)	0.7773	0.9721	0.8727	0.0981	0.9164
SLC-NET [27] (2024)	0.7838	0.9752	0.8780	0.0848	0.9151
HAIC-NET [7] (2024)	0.7816	0.9744	0.8762	**0.0695**	0.9075
Ours	**0.8026**	**0.9762**	**0.8891**	0.0902	**0.9281**

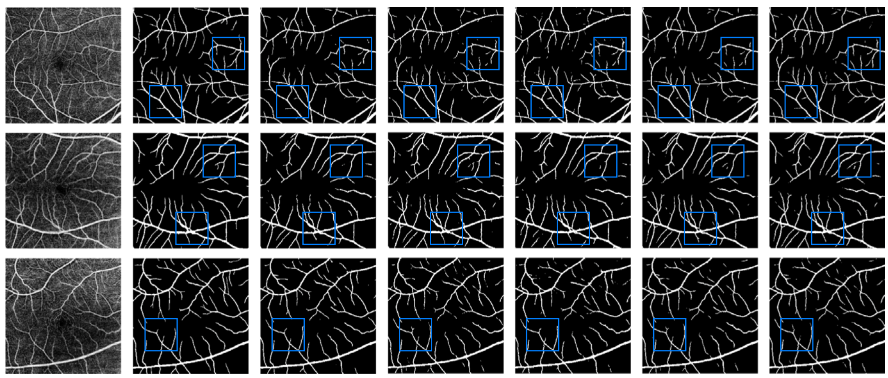


Fig. 6. Visualisation results on ROSSA with only 5% manual annotations. The blue box indicates where DSDC-NET outperforms other network segmentation.

in DSC, and 2.06% in G-Mean over the current state-of-the-art method, HAIC-NET, while maintaining a low FDR. These results highlight the ability of DSDC-NET to improve segmentation accuracy and reduce false positives.

Fig. 6 displays a selection of the visual segmentation results, showing that our method accurately captures the ring structure between vessels, even in complex vessel distributions.

4.5. Ablations

4.5.1. Impact of different annotation data ratios

We validate the effect of annotation ratios on performance using the ROSE-1 dataset. Fig. 7 presents the performance curves of DSDC-NET at various annotation ratios, compared with two other networks designed for OCTA vessel segmentation, DCSS-NET and HAIC-NET, across JAC, DSC, and FDR metrics.

Across different annotation ratios, DSDC-NET consistently achieves superior performance, particularly when the proportion of labeled images is low. Although performance tends to plateau as the proportion increases, DSDC-NET exhibits a more substantial boost than the other networks, indicating that the DTC effectively utilises hidden information in the labeled data. This enables the network to learn more relevant vessel features. As shown in Fig. 7(c), DSDC-NET also reduces false positives and attains state-of-the-art performance as the annotation ratio increases. This suggests that DSAM and DTC enhance image information capture and improve the learning of vessel branching patterns, thus further increasing segmentation accuracy.

4.5.2. Impact of network components

To verify the effectiveness of DSAM and DTC, we conduct experiments on ROSE-1 with a 10.00% annotation ratio. The baseline uses only the classic U-Net architecture. Table 5 presents the ablation results for DSDC-NET components. Compared to the baseline, the performance improvements from using the Snake-Shaped Convolution module, Spatial Attention module, and Persistent Homology Loss individually are modest. The introduction of DSAM and DTC improves JAC by 1.11%, 2.43%, ACC by 0.57%, 1.15%, DSC by 0.85%, 1.84%, and reduces

Table 5
Ablation Study of DSDC-NET Components. The bolded text indicates the optimal component combination. SSC denotes snake-shaped convolution, while SA represents spatial attention mechanism, which forms the Dynamic Spatial Attention Mechanism. ph-loss refers to Persistent Homology loss applied alongside Dice loss to the segmentation results, whereas dice-loss indicates the use of Dice loss on both the skeleton and segmentation results. Collectively, these constitute the dual topological consistency loss.

DSAM		DTC		Metrics				
SSC	SA	ph-loss	dice-loss	JAC↑	ACC↑	DSC↑	FDR↓	G-Mean↑
				0.5998	0.9055	0.7487	0.2214	0.8290
√				0.6035	0.9020	0.7514	0.2543	0.8440
	√			0.6028	0.9055	0.7510	0.2243	0.8322
		√		0.6002	0.9024	0.7491	0.2496	0.8397
			√	0.6197	0.9103	0.7639	0.2150	0.8425
√	√			0.6109	0.9112	0.7572	0.1829	0.8247
		√	√	0.6241	0.9170	0.7671	0.1513	0.8248
√	√	√	√	**0.6433**	**0.9211**	**0.7811**	**0.1452**	**0.8361**

FDR by 3.85%, 7.01%, respectively. The combined use of both modules further enhances performance, validating their effectiveness. Specifically, when integrated with DSAM, DTC increases JAC and DSC by 3.24%, and 2.39%. In contrast, without DSAM, DTC only improves JAC and DSC by 2.43% and 1.84%. These results underscore the synergy between DTC and DSAM's precise vessel localisation, which improves the model's ability to learn branching patterns. Furthermore, combining DSAM with DTC leads to a more substantial performance gain than using DSAM alone, suggesting that when the network learns more stable topological features, DSAM improves vessel detail localisation accuracy, thus enhancing its ability to handle local deformations and complex structures.

Fig. 8 shows the visual results, highlighting that incorporating DSAM and DTC reduces false positives in segmentation. This indicates that DSAM and DTC mitigate false positives by accurately localising vessel details and learning the vessel branching pattern, thereby improving segmentation accuracy.

4.5.3. Impact of topological loss composition

To examine the impact of different topological loss compositions on network performance, we conduct experiments on ROSE-1 with an

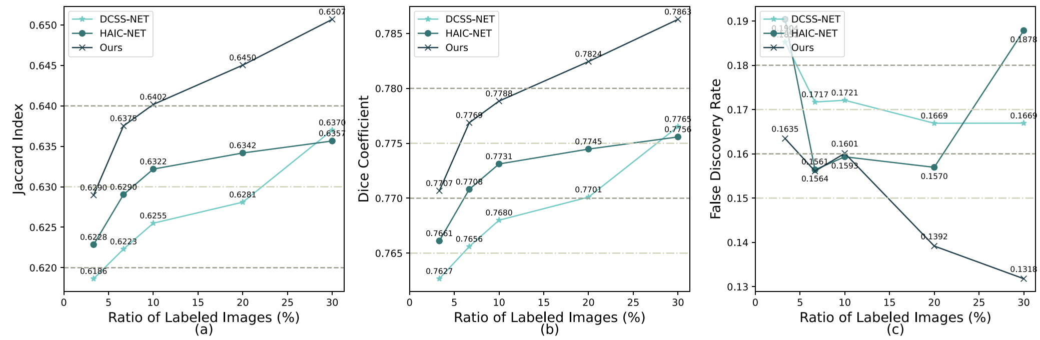
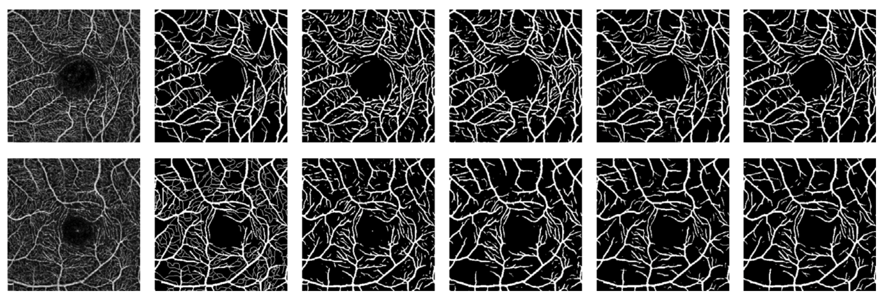
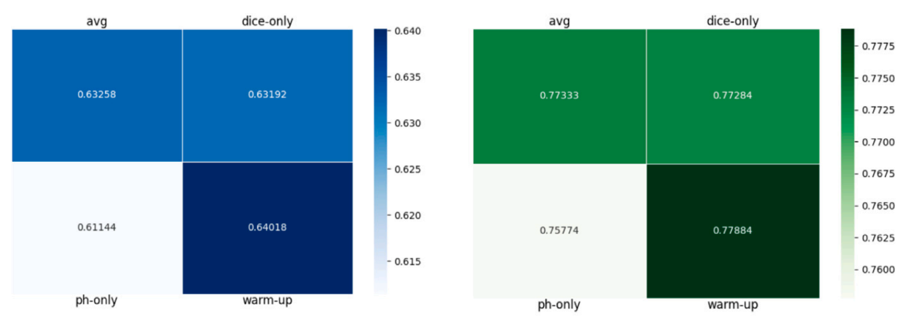


Fig. 7. Network performance under different annotation ratios. (a) Performance curves for the JAC metric; (b) Performance curves for the DSC metric; (c) Performance curves for the FDR metric.



(a) Image (b) GroundTruth (c) baseline (d) baseline+DSAM (e) baseline+DTC (f) DSDC-NET

Fig. 8. Visualisation of the Ablation Study on the Impact of Network Components.



(a) JAC (b) DSC

Fig. 9. Network performance under different topological loss combinations. $dice-only$ refers to the use of Dice loss based solely on the segmentation results and topological skeleton; $ph-only$ denotes Persistent Homology Loss applied only to the segmentation results; avg indicates the arithmetic mean of Dice loss and Persistent Homology Loss; and $warm-up$ represents the weighted average of both losses using the Warm-up mechanism. (a) JAC results for different topological loss combinations; (b) DSC results for different topological loss combinations.

annotation ratio of 10.00%, using various combinations of topological losses. Fig. 9 illustrates the performance variations of DSDC-NET. We evaluated four combinations based on JAC and DSC metrics: (1) Only L_{Dice} ($dice-only$); (2) Only L_{PH} ($ph-only$); (3) Arithmetic mean of L_{Dice} and L_{PH} (avg); and (4) Weighted average using the Warm-up mechanism ($warm-up$).

The results in Fig. 9 show that $ph-only$ performs significantly worse than other combinations, highlighting the limited impact of branch point distribution learning without accurate major vessel segmentation. The avg scenario improves on $dice-only$, with JAC increasing by 0.066% and DSC by 0.049%. The $warm-up$ scenario further enhances performance, with JAC improving by 0.826% and DSC by 0.6% compared to $dice-only$. These results support our design, which prioritises accurate major vessel segmentation before focusing on branching features. L_{Dice} ensures correct segmentation of major vessels, while L_{PH} aids in learning branching patterns. Effective segmentation requires the network to first focus on major vessel features and then learn branching patterns based on this foundation, emphasising the need for varying loss preferences throughout different training stages.

4.5.4. Impact of temperature coefficient τ

We investigate the impact of the temperature coefficient through experiments on the ROSE-1 dataset with a 3.33% annotation ratio over 200 epochs. Fig. 10 shows that network performance improves with increasing temperature coefficients, peaking at $\tau = 21$ before declining. As detailed in Section 4.5.3, the network prioritises learning at different stages. Initially, L_{Dice} emphasises major vessel connectivity and morphology, forming a foundation for later branch segmentation. In later stages, L_{PH} enhances branch point and complex structure identification. Adjusting the temperature coefficient tailors the learning focus: a low coefficient hinders major vessel learning, impacting branch segmentation, while a high coefficient limits branch vessel capture. Thus, selecting the optimal temperature coefficient is crucial.

5. Conclusion

In this paper, we propose an effective Dynamic Spatial Semi-Supervised Vessel Segmentation with Dual Topological Consistency

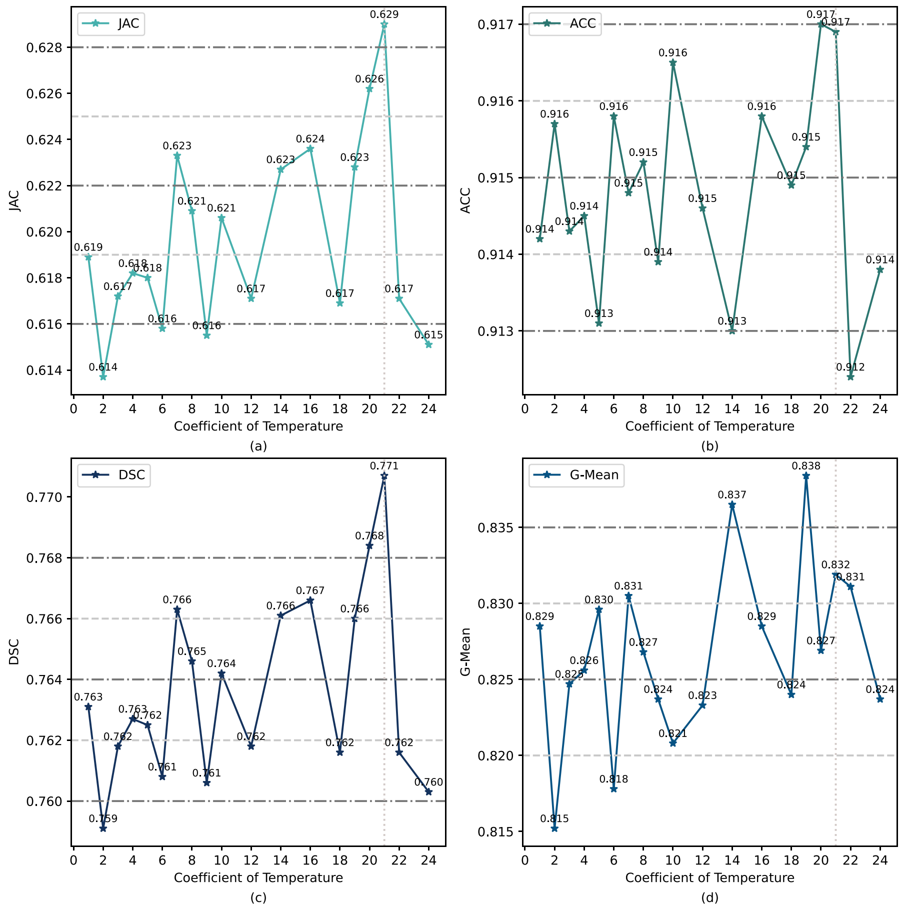


Fig. 10. Network performance under different temperature coefficients. (a) JAC performance variation with varying temperature coefficients; (b) ACC performance variation with varying temperature coefficients; (c) DSC performance variation with varying temperature coefficients; (d) G-Mean performance variation with varying temperature coefficients.

(DSDC-NET). By achieving precise localisation of vessel details and learning the branching pattern of vessels, DSDC-NET significantly enhances its ability to identify fine vessels, reduces false positives, and improves vessel segmentation accuracy. Specially, we combine vessel details with spatial information, enabling accurate localisation of vessel features by perceiving geometric deformations and complex structures while suppressing background noise and artefact interference. Additionally, we improve segmentation connectivity by learning topological features from the entire vessel system and major vessels in OCTA images, while guiding the network to focus on different vessel components during training, thereby enhancing its ability to learn the branching pattern and reduce false positives. Extensive experiments on three datasets demonstrate the efficacy of our method, especially its ability to reduce false positives.

While the results of DSDC-NET are promising, there is still room for improvement in the following areas: (1) Utilisation of Unlabeled Data. Unlabeled data contains valuable information, and enhancing the network ability to extract this information could improve performance. We will develop a supplementary module for vessel learning from unlabeled images, focusing on relationships between major and branch vessels, learned from annotated data, to boost segmentation accuracy. (2) The Physiology of Retinal Vessels. Retinal images contain valuable physiological information that is often underutilised. For example, vessel segmentation should align with physiological flow relationships at branching points. We will design a loss function incorporating abnormal blood flow distribution at these points in OCTA images to enhance segmentation accuracy. (3) Signal Attenuation on Deeper Images. OCTA images at different depths exhibit varying signal attenuation as light passes through tissue layers. While DSDC-NET performs well in superficial OCTA images with stronger signals, its performance may decline in deeper images with greater attenuation. In such cases, we will combine frequency domain filtering with reinforcement learning, allowing the algorithm to dynamically adjust based on the feature distribution in unlabeled images. Feedback from labeled images will guide the network to retain vessel-related information, improving signal quality.

CRediT authorship contribution statement

Xinyi Liu: Writing – review & editing, Writing – original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation. **Hailan Shen:** Supervision, Resources, Funding acquisition. **Wenyan Zhong:** Supervision, Project administration, Conceptualization. **Wanqing Xiong:** Visualization, Project administration, Methodology. **Zailiang Chen:** Supervision, Project administration, Investigation, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 61972419) and the Natural Science Foundation of Hunan Province of China (No. 2025JJ50369).

Data availability

Data will be made available on request.

References

[1] I. Gusar, S. Čanović, M. Ljubičić, S. Šare, S. Perić, I. Caktaš, P. Kovačević, Z. Paštar, S. Konjevoda, Religiousness, anxiety and depression in patients with glaucoma, age-related macular degeneration and diabetic retinopathy, Psychiatr. Danub. (2022).

[2] Y. Liu, L. Zuo, Y. He, S. Han, J. Lei, J.L. Prince, A. Carass, OCTA segmentation with limited training data using disentangled representation learning, in: Deep Learning for Medical Image Analysis, Elsevier, 2024, pp. 451–469.

[3] S. Pedemonte, T. Tsue, B. Mombourquette, Y.N. Truong Vu, T. Matthews, R. Morales Hoil, M. Shah, N. Ghare, N. Zingman-Daniels, S. Holley, et al., A semiautonomous deep learning system to reduce false positives in screening mammography, Radiol. Artif. Intell. 6 (3) (2024) e230033.

[4] J. Zhang, Y. Zhang, H. Qiu, T. Wang, X. Li, S. Zhu, M. Huang, J. Zhuang, Y. Shi, X. Xu, Constrained multi-scale dense connections for biomedical image segmentation, Pattern Recognit. 158 (2025) 111031.

[5] Y. Qi, Y. He, X. Qi, Y. Zhang, G. Yang, Dynamic snake convolution based on topological geometric constraints for tubular structure segmentation, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 6070–6079.

[6] Z. Chen, Y. Xiong, H. Wei, R. Zhao, X. Duan, H. Shen, Dual-consistency semi-supervision combined with self-supervision for vessel segmentation in retinal OCTA images, Biomed. Opt. Express 13 (5) (2022) 2824–2834.

[7] H. Shen, Z. Tang, Y. Li, X. Duan, Z. Chen, HAIC-NET: Semi-supervised OCTA vessel segmentation with self-supervised pretext task and dual consistency training, Pattern Recognit. 151 (2024) 110429.

[8] S. Trivedi, U. Sharma, M. Rathore, M. John, Comprehensive study of arrangement of renal hilar structures and branching pattern of segmental renal arteries: An anatomical study, Cureus (2023).

[9] O. Ronneberger, P. Fischer, T. Brox, U-net: Convolutional networks for biomedical image segmentation, in: International Conference on Medical Image Computing and Computer-Assisted Intervention, 2015.

[10] Y. Ma, H. Hao, J. Xie, H. Fu, J. Zhang, J. Yang, Z. Wang, J. Liu, Y. Zheng, Y. Zhao, ROSE: a retinal OCT-angiography vessel segmentation dataset and new model, IEEE Trans. Med. Imaging 40 (3) (2020) 928–939, http://dx.doi.org/10.1109/TMI.2020.3042802.

[11] Y. Liu, A. Carass, L. Zuo, Y. He, S. Han, L. Gregori, S. Murray, R. Mishra, J. Lei, P.A. Calabresi, et al., Disentangled representation learning for OCTA vessel segmentation with limited training data, IEEE Trans. Med. Imaging 41 (12) (2022) 3686–3698.

[12] G. Cao, Z. Peng, Z. Zhou, Y. Wu, Y. Zhang, R. Yan, Multi-task OCTA image segmentation with innovative dimension compression, Pattern Recognit. 159 (2025) 111123.

[13] D. Berthelot, N. Carlini, I. Goodfellow, N. Papernot, A. Oliver, C.A. Raffel, Mixmatch: A holistic approach to semi-supervised learning, Adv. Neural Inf. Process. Syst. 32 (2019).

[14] A. Tarvainen, H. Valpola, Mean teachers are better role models: Weight-averaged consistency targets improve semi-supervised deep learning results, Adv. Neural Inf. Process. Syst. 30 (2017).

[15] C. Li, W. Ma, L. Sun, X. Ding, Y. Huang, G. Wang, Y. Yu, Hierarchical deep network with uncertainty-aware semi-supervised learning for vessel segmentation, Neural Comput. Appl. (2022) 1–14.

[16] N. Shen, T. Xu, S. Huang, F. Mu, J. Li, Expert-guided knowledge distillation for semi-supervised vessel segmentation, IEEE J. Biomed. Heal. Informatics (2023).

[17] I. Misra, L.v.d. Maaten, Self-supervised learning of pretext-invariant representations, in: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2020, pp. 6707–6717.

[18] L. Zhao, J. Gao, D. Deng, X. Li, SSIR: Spatial shuffle multi-head self-attention for single image super-resolution, Pattern Recognit. 148 (2024) 110195.

[19] Q. Truong, P. Chin, Weisfeiler and lehman go paths: Learning topological features via path complexes, in: Proceedings of the AAAI Conference on Artificial Intelligence, Vol. 38, No. 14, 2024, pp. 15382–15391.

[20] X. Luo, L. Peng, Z. Ke, J. Lin, Z. Yu, PA-net: A hybrid architecture for retinal vessel segmentation, Pattern Recognit. 161 (2025) 111254, URL https://www.sciencedirect.com/science/article/pii/S0031320324010057.

[21] N. Tajbakhsh, L. Jeyaseelan, Q. Li, J.N. Chiang, Z. Wu, X. Ding, Embracing imperfect datasets: A review of deep learning solutions for medical image segmentation, Med. Image Anal. 63 (2020) 101693.

[22] M. Li, Y. Zhang, Z. Ji, K. Xie, S. Yuan, Q. Liu, Q. Chen, OCTA-500: A retinal dataset for optical coherence tomography angiography study, Med. Image Anal. 93 (2024) 103092, http://dx.doi.org/10.1016/j.media.2024.103092.

[23] H. Ning, C. Wang, X. Chen, S. Li, An accurate and efficient neural network for octa vessel segmentation and a new dataset, in: ICASSP 2024-2024 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE, 2024, pp. 1966–1970, http://dx.doi.org/10.1109/ICASSP48485.2024.10447708.

[24] A. Kirillov, E. Mintun, N. Ravi, H. Mao, C. Rolland, L. Gustafson, T. Xiao, S. Whitehead, A.C. Berg, W.-Y. Lo, et al., Segment anything, in: Proceedings of the IEEE/CVF International Conference on Computer Vision, 2023, pp. 4015–4026.

[25] X. Li, L. Yu, H. Chen, C.-W. Fu, P.-A. Heng, Semi-supervised skin lesion segmentation via transformation consistent self-ensembling model, 2018, arXiv preprint arXiv:1808.03887.

[26] X. Li, L. Yu, H. Chen, C.-W. Fu, L. Xing, P.-A. Heng, Transformation-consistent self-ensembling model for semisupervised medical image segmentation, IEEE Trans. Neural Networks Learn. Syst. 32 (2) (2020) 523–534.

[27] J. Liu, C. Desrosiers, D. Yu, Y. Zhou, Semi-supervised medical image segmentation using cross-style consistency with shape-aware and local context constraints, IEEE Trans. Med. Imaging 43 (4) (2024) 1449–1461.

[28] C. Wang, H. Ning, X. Chen, S. Li, Db-unet: Mlp based dual branch unet for accurate vessel segmentation in octa images, in: ICASSP 2023-2023 IEEE International Conference on Acoustics, Speech and Signal Processing, ICASSP, IEEE, 2023, pp. 1–5.

Xinyi Liu received a Bachelors degree in data science and big data technology from Fuzhou University in 2023. Since then, she has moved on to pursue her study for a Masters degree at the School of Computer Science and Engineering, Central South University.

Hailan Shen received a Ph.D. degree in computer science from Central South University in 2011. She is currently an associate professor in the School of Computer Science and Engineering at Central South University, China. Her research interests include Medical Image Analysis, Machine learning, and Artificial Intelligence.

Wenyan Zhong received a Bachelors degree in software engineering from Nanchang University in 2023. Since then, she has moved on to pursue her study for a Masters degree at the School of Computer Science and Engineering, Central South University.

Wanqing Xiong received a Bachelors degree in software engineering from Nanjing University of Aeronautics and Astronautics in 2022. Since then, she has moved on to pursue her study for a Masters degree at the School of Computer Science and Engineering, Central South University.

Zailiang Chen received a Ph.D. degree in computer science from Central South University in 2012. He is currently an associate professor in the School of Computer Science and Engineering, Central South University. His research interests include Computer Vision, Machine learning, Medical Image Analysis and Artificial Intelligence.